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B.Tech. Degree VIII Semester Examination April 2017

ME 1801 COMPRESSIBLE FLUID FLOW (2012 Scheme)

(Use of approved gas tables allowed)

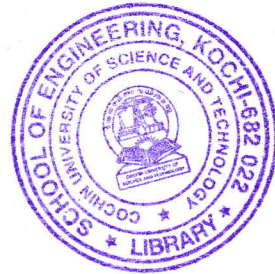
Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) What is International Standard Atmosphere? Derive an expression to determine the pressure variation in the Troposphere.
- (b) Derive the differential form of continuity equation for compressible flows.
- (c) Why are normal shocks impossible in subsonic flows? Explain.
- (d) Derive the Prandtl's equation for a flow through an oblique shock.
- (e) What is meant by friction choking? Explain.
- (f) Derive the slope of Rayleigh curve on h-s and p-v diagrams.
- (g) Explain the working principle of Prandtl Pitot static tube with a neat diagram.
- (h) Explain the working principle of shock tube with a neat diagram.



PART B

(4 × 15 = 60)

- II. (a) Derive the area – velocity relationship for accelerating and decelerating passages for various values of Mach numbers. (5)
 - (b) An aircraft is flying at an altitude of 12 km with a Mach number of 0.82. The cross sectional area of the inlet diffuser is 0.5 m^2 . Determine (i) the mass flow rate of air entering the diffuser (ii) the speed of aircraft (iii) stagnation pressure and temperature of air at the diffuser entry. (10)
- OR**
- III. (a) Describe the behavior of flow in a convergent-divergent nozzle when it is operated at (i) design pressure ratio (ii) pressure ratio higher than the design value (iii) pressure ratio lower than the design value. (5)
 - (b) Air flows through a C-D nozzle which has inlet area of 10 cm^2 . The air has a velocity of 80 m/s, a temperature of 301 K and a pressure of 700 kPa at the inlet section. The pressure at the exit section is 250 kPa. Determine the mass flow rate and the velocity of air at the exit section of the nozzle. Assume on dimensional isentropic flow through the nozzle. (10)
- IV. (a) What are the characteristics of Mach waves? (5)
 - (b) Air enters a C-D nozzle of throat area 10 cm^2 with a pressure of 2.9 MPa and a temperature of 323 K. A normal shock occurs in the downstream section where the pressure is 0.5 MPa. Determine the Mach number and pressure just behind the shock wave. Also determine the mass flow rate through the nozzle. (10)

OR

(P.T.O.)

- V. Air flowing with a Mach number of 2 with a pressure of 80 kPa and a temperature of 303 K passes over a component of an aircraft that can be modeled as a wedge with an included angle of 8° that is aligned with the flow. The flow is turned through an angle of 4° , leading to the generation of oblique shock waves on both sides. Determine the pressure acting on the surface of the wedge. (15)
- VI. Air flows out of a pipe of diameter 0.3 m at a rate of $1000 \text{ m}^3/\text{minute}$ with a pressure and temperature of 150 kPa and 293 K respectively. If the pipe is 50 m long and assuming $f = 0.005$, determine the Mach number at the exit, the inlet pressure and the inlet temperature. (15)
- OR**
- VII. (a) What is meant by thermal choking? Explain. (5)
 (b) The pressure, temperature and velocity of a gas at the entry to a combustor are 0.35 bar, 310 K and 60 m/s respectively. Determine the Mach number, pressure and temperature at the exit if the increase in stagnation enthalpy of the gas between entry and exit is 1200 kJ/kg. Take $C_p = 1005 \text{ J/kg.K}$ and $\gamma = 1.4$. (10)
- VIII. Explain the following terms. (15)
 (i) Rayleigh Supersonic Pitot formula (ii) Temperature recovery factor
 (iii) Starting and steady operation of supersonic wind tunnels.
- OR**
- IX. Explain the working of the following with the help of neat figures. (15)
 (i) Schlieren apparatus (ii) Interferometer (iii) Velocimeter
 (iv) Hot wire anemometer (v) Shadowgraph.
